

SEMESTER END EXAMINATIONS – JULY / AUGUST 2024

Program	: B.E. –Common to MECH / IEM /CHE	Semester	: II
Course Name	: Multivariate Calculus and Differential Equations	Max. Marks	: 100
Course Code	: MAM21	Duration	: 3 Hrs

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

- State Taylor's series for the function of two variables. CO1 (02)
 - Solve the system of non-linear equations $x^2 + y^2 = x$, $x^2 - y^2 = y$, given that $x_0 = 0.8$ and $y_0 = 0.4$ using Newton-Raphson method (Carry out one iteration only). CO1 (04)
 - Using Maclaurin's series expand $\sqrt{1 + \sin 2x}$ up to the term containing x^4 . CO1 (07)
 - A rectangular box open at the top is to have volume of 108 cubic ft. Find the dimension of the box if its total surface area is minimum. CO1 (07)
- State Maclaurin's series for the function of one variable. CO1 (02)
 - Show that $f(x, y) = xy(1 - x - y)$ is maximum at the point $(1/3, 1/3)$. CO1 (04)
 - Expand the function $xy^2 + \cos(xy)$ about $(1, \pi/2)$ up to second degree terms. CO1 (07)
 - Find a real root of the equation $x \sin x + \cos x = 0$ correct to three decimal places using Newton-Raphson method. CO1 (07)

UNIT - II

- Write Euler's formula to solve first order and first degree ODE. CO2 (02)
 - Find the orthogonal trajectory of $r = b \sin \theta$. CO2 (04)
 - If the temperature of the air is 30°C and the substance cools from 100°C to 70°C in 15 minutes, find when the temperature will be 40°C . CO2 (07)
 - Solve $\frac{dy}{dx} = xy + y^2$; $y(0) = 1$ at $x = 0.2$ taking $h = 0.2$ using Runge-Kutta method of fourth order. CO2 (07)
- Define orthogonal trajectories. CO2 (02)
 - Using Taylor's series method solve $\frac{dy}{dt} = e^{-2t} - 2y$; $y(1) = 0.1$ at $t = 1.1$ by considering upto third degree term. CO2 (04)
 - Solve $\frac{dy}{dx} = x + \sqrt{y}$; $y(0) = 1$ at $x = 0.2$ taking $h = 0.2$ using Modified Euler's method. CO2 (07)

- d) A voltage Ee^{-at} is applied at $t = 0$ to a circuit of inductance L and resistance R . Show that the current at time t is $\frac{E}{R - aL} \left(e^{-at} - e^{-\frac{Rt}{L}} \right)$ if initial current is zero. CO2 (07)

UNIT - III

5. a) Write the complementary function of the fourth order LDE $f(D)y = 0$, if two roots of the auxiliary equation are imaginary and two roots are real and distinct. CO3 (02)
- b) Solve the initial value problem $y'' - y = 0$, $y(0) = 3$, $y'(0) = -3$; CO3 (04)
- c) Solve $(D^2 - 2D + 4)y = e^x \sin x$. CO3 (07)
- d) Solve $\frac{d^2 y}{dx^2} + \frac{1}{x} \frac{dy}{dx} = \frac{12 \log x}{x^2}$. CO3 (07)
6. a) Define boundary value problem. CO3 (02)
- b) Find the P.I. of the differential equation $y'' + y' = x^2 + 2x + 4$. CO3 (04)
- c) Solve $y'' - 6y' + 9y = \frac{e^{3x}}{x^2}$ by the method of variation of parameters. CO3 (07)
- d) Solve $(2x + 1)^2 y'' - 2(2x + 1)y' - 12y = 6x + 5$. CO3 (07)

UNIT-IV

7. a) Write the Simpson's $1/3^{rd}$ rule for numerical integration. CO4 (02)
- b) Use Lagrange's interpolation formula to find the value of $f(1)$ from the following data: CO4 (04)

x	-1	0	2
$f(x)$	9	5	3

- c) Find $f(5)$ from the following data by using Newton divided difference formula. CO4 (07)

x	0	1	3	4
$f(x)$	-4	1	29	52

- d) Find $y'(0)$ and $y''(0)$ from the following table: CO4 (07)

x	0	2	4	6
$f(x)$	0	12	248	1284

8. a) Write Lagrange's inverse interpolation formula. CO4 (02)
- b) Find the missing terms from the following data CO4 (04)

x	45	50	55	60	65
y	3.0	-	2.0	-	-2.4

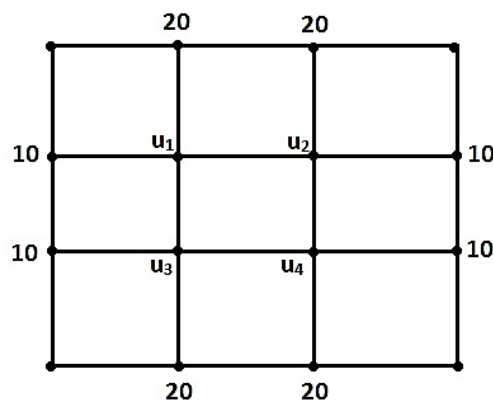
- c) In a medical diagnosis test the number of persons who tested Covid positive between different age groups is as follows. Estimate the number of Covid infected persons between the age group 35 and 40. CO4 (07)

Age group	20 – 30	30 – 40	40 – 50	50 – 60	60 – 70
No. of persons infected	20	35	58	40	28

- d) Evaluate $\int_0^1 \frac{dx}{(1+x)^2}$ using Simpson's 3/8th rule taking 6 subintervals. CO4 (07)

UNIT - V

9. a) Write Bendre-Schmidt recurrence relation to solve one dimensional heat equation. CO5 (02)
- b) Classify the partial differential equation $2 \frac{\partial^2 z}{\partial x^2} + 4 \frac{\partial^2 z}{\partial x \partial y} + 3 \frac{\partial^2 z}{\partial y^2} = 2 \cdot$ CO5 (04)
- c) Solve $\frac{\partial z}{\partial x} = 2 \frac{\partial z}{\partial y} + z$, $z(x,0) = 6e^{-3x}$ using the method of separation of variables. CO5 (07)
- d) Perform two iterations of Gauss-Seidel method to find approximate solution of the equation $u_{xx} + u_{yy} = 0$ for the square mesh with boundary values as shown in figure. CO5 (07)



10. a) Write Standard five point formula. CO5 (02)
- b) Solve the PDE $\frac{\partial^2 u}{\partial x \partial y} = \frac{x}{y} + a$, by direct integration method. CO5 (04)
- c) Solve the PDE $\frac{y^2 z}{x} p + xz q = y^2$ by Lagrange's method. CO5 (07)
- d) Solve $u_{tt} = 5u_{xx}$ with the boundary conditions $u(0,t) = 0$; $u(6,t) = 0$ and the initial conditions $u(x,0) = x(x+3)$, $0 < x < 6$ and $u_t = 0$ at $t = 0$. Compute u for two time levels taking $h=1$ & $k = 0.5$. CO5 (07)
